Exp. No: 1

Transformation between Cartesian, Cylindrical and Spherical Coordinate Systems

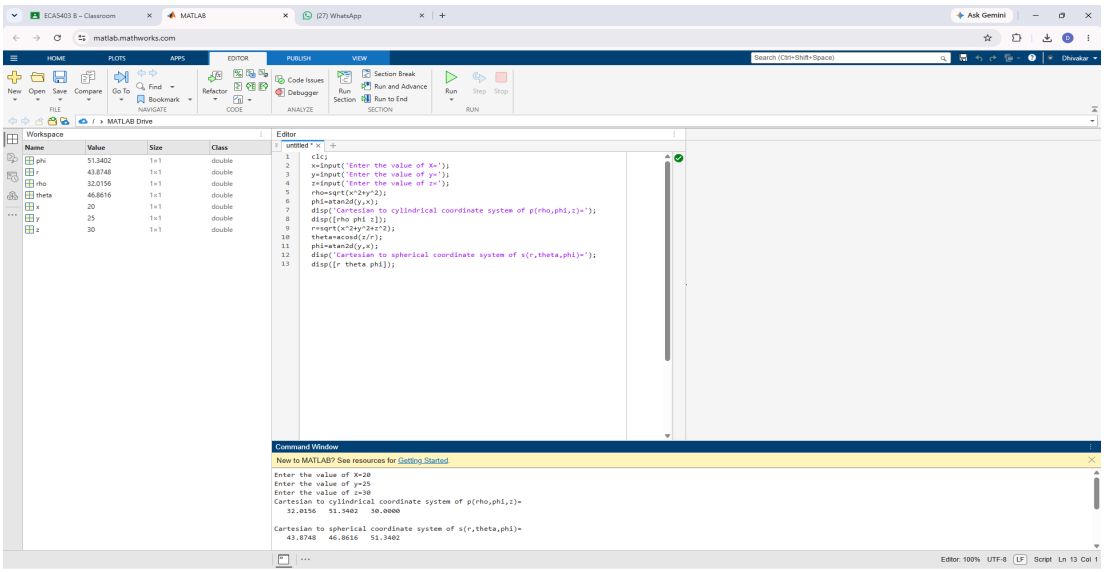
Objectives

To understand the mathematical relationships between Cartesian, Cylindrical and Spherical coordinate systems. To develop a MATLAB program that converts the position of an object from one coordinate system to another. To verify the correctness of the coordinate transformation by comparing computed values against standard analytical formulas.

Test Case (i): Cartesian to Cylindrical and Spherical Coordinates

Input values: x = 10, y = 15, z = 20

```
clc; x=input('Enter the value of X=');
y=input('Enter the value of y=');
z=input('Enter the value of z=');
% Cartesian to Cylindrical rho=sqrt(x^2+y^2);
phi=atan2d(y,x);
disp('Cartesian to cylindrical coordinate system of p(rho,phi,z)=');
disp([rho phi z]); % Cartesian to Spherical r=sqrt(x^2+y^2+z^2);
theta=acosd(z/r);
phi=atan2d(y,x);
disp('Cartesian to spherical coordinate system of s(r,theta,phi)='); disp([r theta phi]);
```



Test Case (ii): Cylindrical to Cartesian and Spherical Coordinates

Input values: ρ = 10, ϕ = 15°, z = 20

clc;

rho=input('Enter the value of rho=');

phi=input('Enter the value of phi=');

z=input('Enter the value of z=');

% Cylindrical to Cartesian

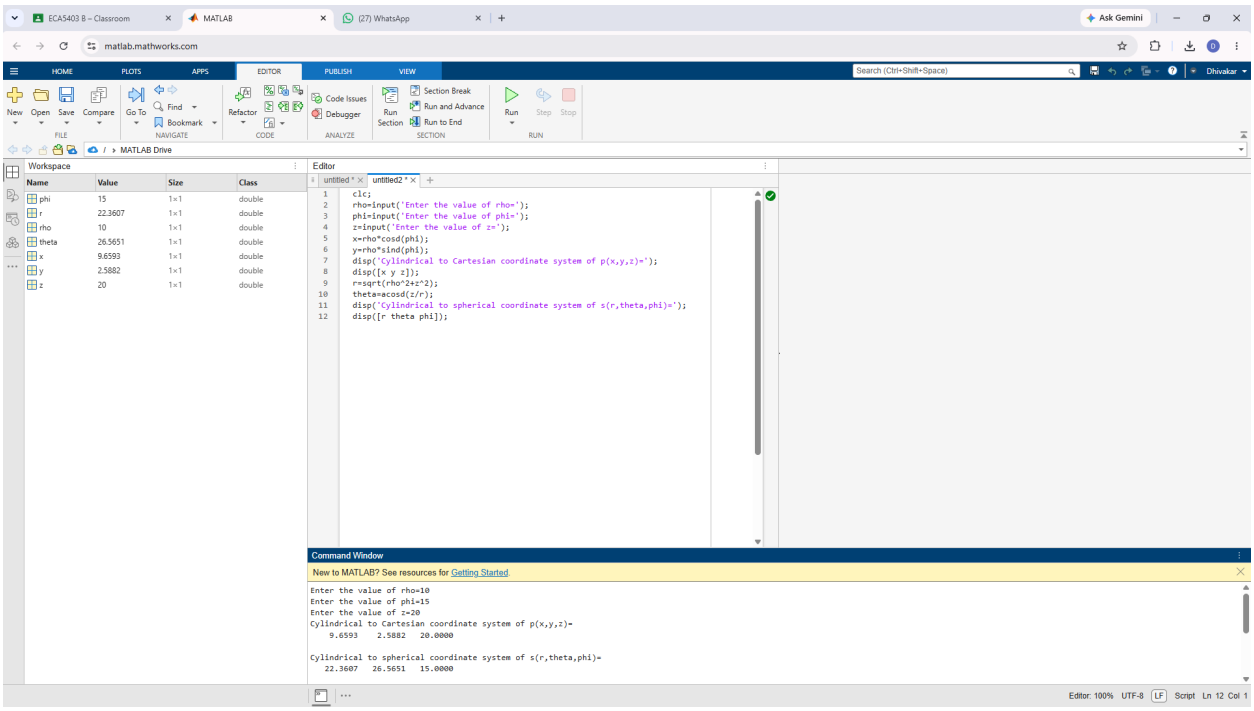
x=rho*cosd(phi); y=rho*sind(phi);

disp('Cylindrical to Cartesian coordinate system of p(x,y,z)='); disp([x y z]);

% Cylindrical to Spherical r=sqrt(rho^2+z^2);

theta=acosd(z/r);

disp('Cylindrical to spherical coordinate system of s(r,theta,phi)=');



Test Case (iii): Spherical to Cartesian and Cylindrical Coordinates Input values: r = 10, θ = 20°, ϕ = 15° (spherical coordinates are entered in the order r, θ, ϕ).

clc;

r=input('Enter the value of r='); theta=input('Enter the value of theta (deg)='); phi=input('Enter the value of phi (deg)=');

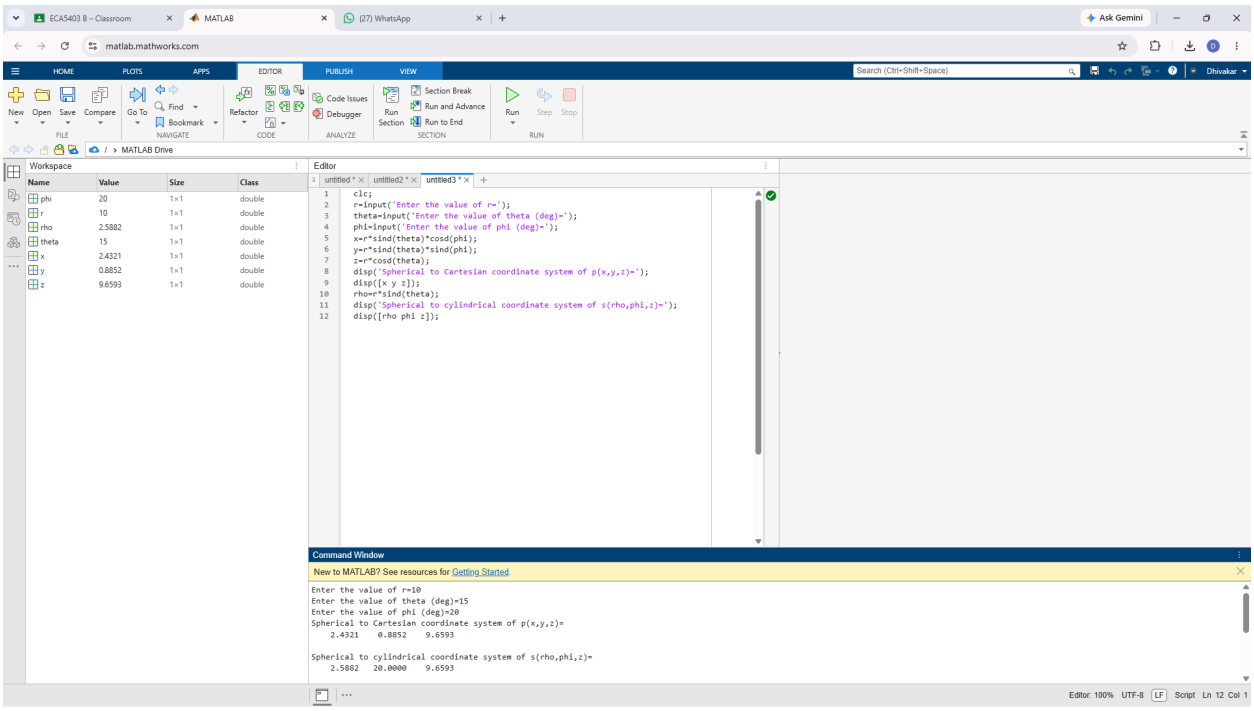
% Spherical to Cartesian x=r*sind(theta)*cosd(phi);

y=r*sind(theta)*sind(phi);

z=r*cosd(theta); disp('Spherical to Cartesian coordinate system of p(x,y,z)=');

disp([x y z]);

% Spherical to Cylindrical rho=r*sind(theta); disp('Spherical to cylindrical coordinate system of s(rho,phi,z)='); disp([rho phi z]);


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MATLAB
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matlab.mathworks.com
HOME
PLOTS
APPS
EDITOR
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Search (Ctrl+Shift+Space)
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Open
Save
Compare
Go To
Find
Bookmark
Refactor
Code Issues
Debugger
Run Section
Section Break
Run and Advance
Run to End
Run
Step
Stop
FILE
NAVIGATE
CODE
ANALYZE
SECTION
RUN
/ > MATLAB Drive
Workspace
Name
Value
Size
Class
phi 20 1×1 double
r 10 1×1 double
rho 2.5882 1×1 double
theta 15 1×1 double
x 2.4321 1×1 double
y 0.8852 1×1 double
z 9.6593 1×1 double
Editor
untitled *
untitled2 *
untitled3 *
1 clc;
2 r=input('Enter the value of r=');
3 theta=input('Enter the value of theta (deg)=');
4 phi=input('Enter the value of phi (deg)=');
5 x=r*sind(theta)*cosd(phi);
6 y=r*sind(theta)*sind(phi);
7 z=r*cosd(theta);
8 disp('Spherical to Cartesian coordinate system of p(x,y,z)=');
9 disp([x y z]);
10 rho=r*sind(theta);
11 disp('Spherical to cylindrical coordinate system of s(rho,phi,z)=');
12 disp([rho phi z]);
Command Window
New to MATLAB? See resources for Getting Started.
Enter the value of r=10
Enter the value of theta (deg)=15
Enter the value of phi (deg)=20
Spherical to Cartesian coordinate system of p(x,y,z)=
2.4321 0.8852 9.6593
Spherical to cylindrical coordinate system of s(rho,phi,z)=
2.5882 20.0000 9.6593
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S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Apply coordinate-system definitions and angle conventions	Apply coordinate-system definitions, symbols, units, and angle conventions	Applies coordinate-system definitions and angle conventions throughout.	Uses correct definitions with one minor notation or convention error.	Shows partial understanding; multiple corrections are needed.	Uses incorrect coordinate definitions or angle conventions.	25	2.5	
2	Implement transformations in MATLAB	Implement six transformations in MATLAB	Implements all six transformations correctly using suitable degree-based functions.	Implements most transformations correctly with minor coding errors.	Implements some transformations but needs assistance to correct errors.	Unable to implement the required transformations.	25	2.5	
3	Verify numerical values and reverse transformations	Verify numerical values and reverse transformations	All outputs agree with analytical values; reverse transformations recover the original point within rounding precision.	Most values agree; small unexplained differences occur.	Several values differ or reverse verification is incomplete.	Outputs are incorrect or not verified.	30	3	
4	Interpret and document results	Interpret and document results	Presents complete tables, units, deviations, and a clear conclusion.	Documentation is clear with minor omissions.	Tables or interpretation are incomplete.	Results and conclusion are missing or unclear.	20	2	
Total							100	10	